

One-step mobile-network traffic forecasting in Milan

Gentil Christian Tonny Iradukunda | ML Techniques I | Formative Assignment 1

Video: <https://youtu.be/6sHUK8ux-5g> · Code: <https://github.com/irachrist1/ml-techniques-1-formative-1>

1. Introduction

This study asks how three distinct sequential models compare for the next ten minutes of Internet activity, and whether their performance varies across Milan areas. What I predict is the dataset's activity count, not bandwidth and not bytes. So every error figure in this report is in activity units, and none of them should be read as megabytes or as a capacity number.

The complete November-December 2013 collection is processed to rank 10,000 areas. RidgeAR, LSTM and CausalCNN are compared with immediate persistence and daily-seasonal persistence. December 16-22 supplies the evaluation week.

2. Related work and model motivation

Barlacchi et al. [1] describe the spatial aggregation and scaling of telecommunications records and their daily and weekly variation. This motivates inspecting each area's temporal structure. No land-use identity is inferred for squares 4159 or 4556: the paper's often-quoted Bocconi and Navigli examples are squares 4259 and 4456, so neither label transfers to the squares this assignment specifies.

Azari et al. [2] compare LSTM and ARIMA on cellular packet traffic; the usefulness of complexity depends on features, training length and traffic conditions. Here, strong lag-one dependence motivates RidgeAR as a low-cost linear comparator. Its L2 penalty stabilizes correlated lag coefficients, but it cannot learn nonlinear interactions. It predicts a correction to persistence and is not ARIMA: there are no moving-average error terms.

LSTM tests whether gated nonlinear processing of a day's observations improves on linear lag weights. Its cell and hidden states evolve within each window, not between windows. This flexibility costs training and inference time; the one-day input cannot expose a complete weekly cycle.

Bai et al. [3] motivate causal dilated convolutions as an alternative to recurrence. CausalCNN combines nonlinear local features across the input window, but omits their residual blocks and weight normalization. Its finite receptive field and padding are explicit below; published benchmark rankings are not assumed to transfer. Zhang and Patras [4] use spatial information and longer horizons, motivating extensions rather than a directly comparable accuracy target.

The reference-seed result on area 5161 favors CausalCNN (RMSE 118.16 versus persistence 134.88). Area and seed comparisons below qualify that result.

3. Dataset, preparation and memory management

The 61 daily files contain 314,966,126 rows and 20.48 GB. January 1 is excluded. Downloads and daily preparation check publisher MD5 values. Timestamps use Europe/Rome boundaries and a ten-minute grid. Square IDs are validated before conversion to int32; activity remains float64.

Country-code Internet values are summed within each square and interval. Empty fields are excluded, observed zeros are retained, and a bin is missing when it has no observed Internet values. There are 145,354 missing bins among 87,840,000. One thing I cannot check from this data: a bin can have some country rows and be missing others, and there is no way to tell that apart from full coverage.

Projected, chunked input and fixed daily sum/count arrays (17.28 MB) avoid retaining the full collection in RAM. Arrays are not the whole process footprint: parser objects, validation and transient allocations also contribute. Raw files remain on disk, trading storage and I/O for bounded processing memory.

Table 1. Repeated November 1 benchmark; decimal MB, medians over 3 isolated processes per method.

Method	Peak RSS	RSS range	Wall seconds	Largest frame
baseline	1087.6	773.5-1125.4	2.32	309.9
chunked	384.7	347.1-391.1	3.78	2.0

The median peak reduction is 64.6%. Every pair matches missing masks and observation counts; maximum aggregate difference is 1.8e-12. Order alternates, but cache and background load are uncontrolled. Chunked processing is the slower of the two here, and most of that gap is the per-chunk square-ID validation rather than the chunking itself: reading the same file with the IDs narrowed straight to int32 runs in about 1.8 s against 3.7 s with validation, so before that check was added the chunked path was marginally faster than the full-day load. I keep the check because narrowing first lets an out-of-range ID wrap into a valid one. Wall time leaves out imports, checksums and writing the output; peak memory includes imports and everything up to that point. An earlier single run of this benchmark reported 86.4%. It did not reproduce, so I keep it in `memory_benchmark.json` for the record but I do not rely on it.

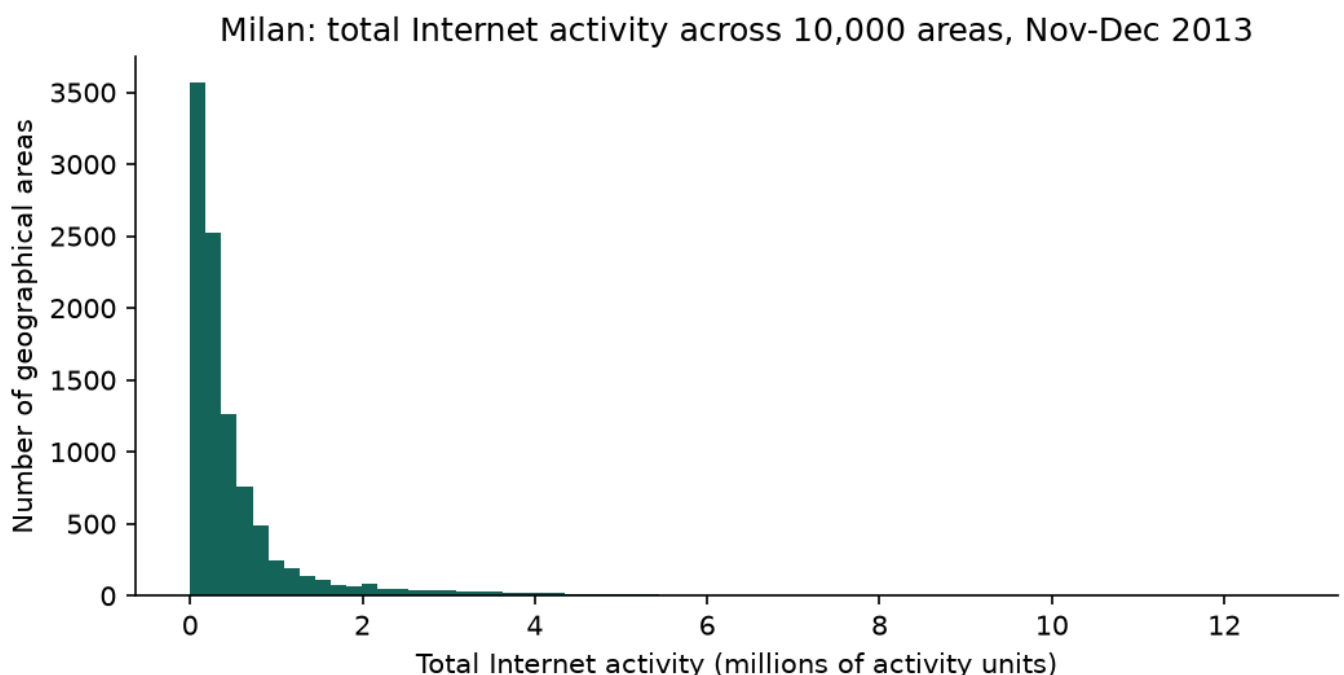


Figure 1. Total observed activity across 10,000 areas over November-December.

The distribution is right-skewed: median 274,706, 99th percentile 4,666,202, maximum 12,682,673. The top three are **5161, 5059, 5259**, holding 0.62% of total observed activity. Under the assumption that missing bins equal each area's observed mean, the largest adjusted partial-area total is 891,902, below third place (10,436,792). That is a what-if under an assumption, not a hard limit on how much traffic could be hiding in the missing bins. I also picked the areas using totals over the whole period, which includes the evaluation week. The brief asks for that, but it does mean my choice of area is not independent of the week I test on.

4. Exploratory and temporal analysis

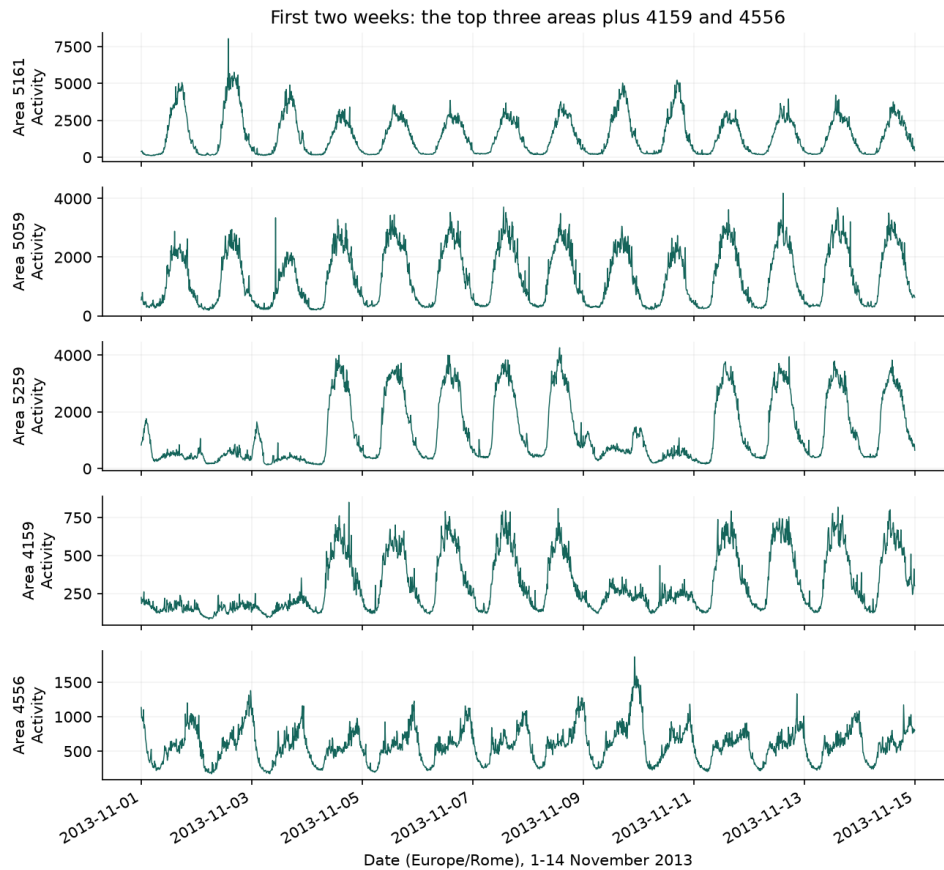


Figure 2. November 1-14 for the top three areas and squares 4159 and 4556; panels retain their own scales.

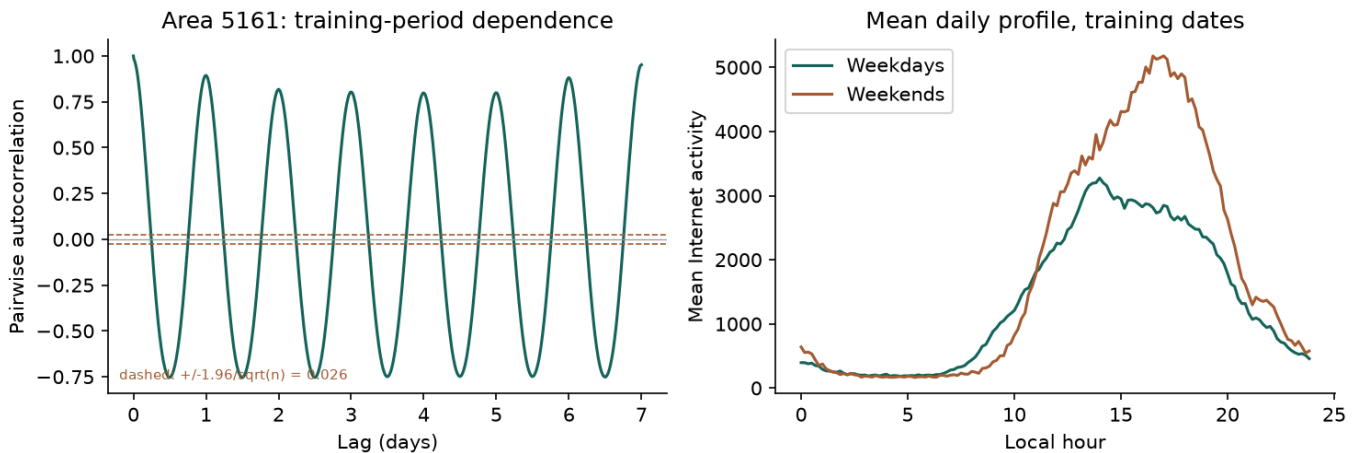


Figure 3. Training-only pairwise autocorrelation and weekday/weekend profiles for area 5161. Dashed lines are a white-noise reference (see reference 9), not confidence intervals for this seasonal series.

Areas 5259 and 4159 have weekday plateaus and reduced weekend activity. Area 5161 has stronger weekend peaks; 5059 has broad daytime plateaus. Area 4556 contains isolated spikes and lower lag-one correlation, but its coefficient of variation is the lowest of the five. One calendar detail matters for reading the left edge of Figure 2: 1 November 2013 was All Saints' Day, a public holiday in Italy, and it fell on a Friday, so the opening low stretch is a three-day holiday weekend and the 9-10 November weekend is the cleaner comparison. These differences suggest different activity schedules without identifying land use or event causes.

The two additional analyses are temporal dependence and weekday/weekend profiles. Lag-one correlation 0.982 motivates persistence and recent-history inputs; daily 0.894 and weekly 0.952 correlations motivate testing longer histories and future calendar features. Six hours is near zero (-0.017); the half-day trough is at twelve hours (-0.753). These are descriptive correlations, not proof that any architecture will forecast best.

4.1 Statistical characterization

Table 2. Training-period characteristics; all five selected full-period series have zero missing bins.

Area	Mean	CV	Lag 1	Lag 144	Weekend/weekday
5161	1511.4	0.92	0.982	0.894	1.36
5059	1342.6	0.71	0.972	0.912	0.81
5259	1314.1	0.85	0.989	0.669	0.42
4159	317.2	0.59	0.969	0.743	0.54
4556	582.4	0.43	0.943	0.769	1.14

CV is sample standard deviation divided by mean. Area 5161 averages 1847 on weekends and 1357 on weekdays, whereas the weekend/weekday ratio is 0.42 in area 5259. These profiles support separate area-level evaluation. A repeated daily shape need not make yesterday's level a good ten-minute forecast.

On 5,472 complete training values, ADF [8] gives statistic -4.103, $p=0.000957$. The regression includes a constant; AIC selects 152 lags from a maximum of 168. Expanding the previous 30-lag search changed p from $1.35e-27$ to $9.57e-04$, demonstrating specification sensitivity. The expanded candidate set permits daily-lag dependence; it is not a universal rule that ADF must include a full seasonal cycle. Both specifications reject the unit-root null under their assumptions. Neither proves strict stationarity or removes seasonal structure; residual diagnostics and further specification checks remain limitations. First-differenced $p=3.3e-15$ does not by itself justify differencing the forecasting target.

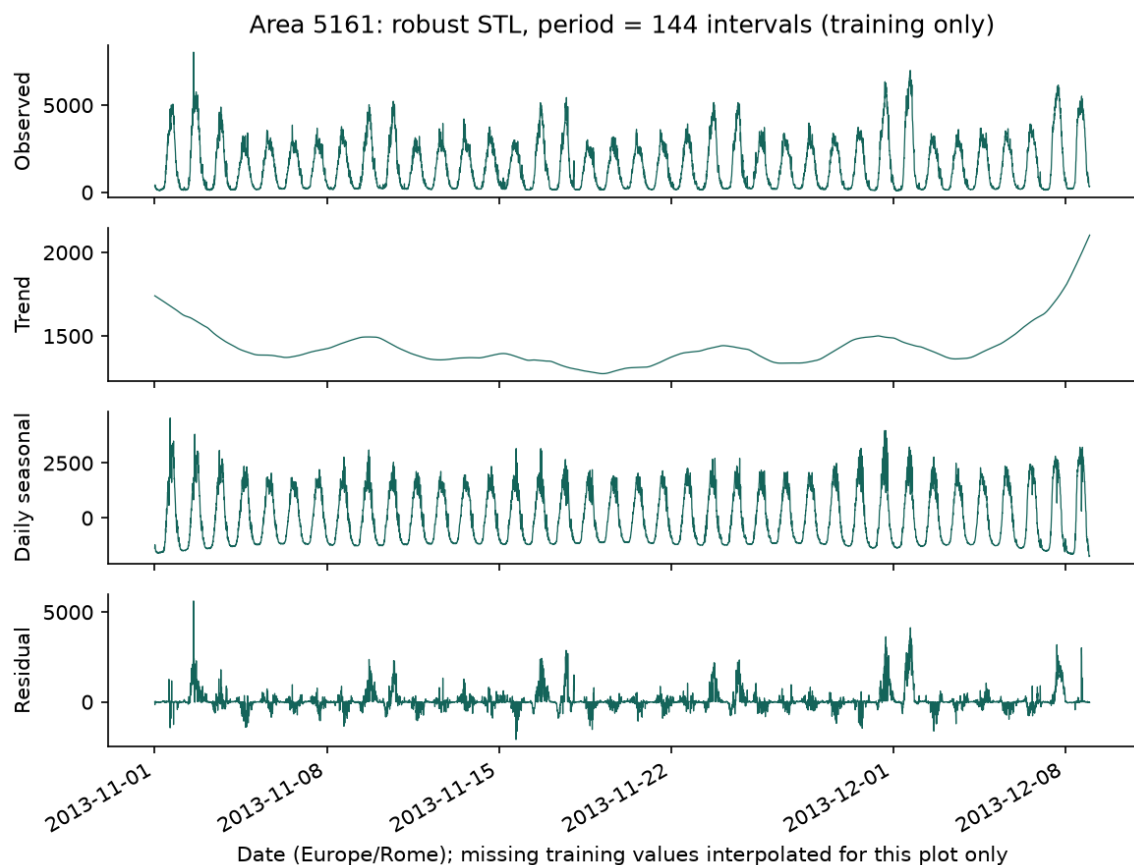


Figure 4. Robust STL (see reference 10) on training dates, period 144. The daily seasonal component, changing trend and residual spikes describe different sources of variation.

Daily seasonal strength is 0.870, defined as $\max(0, 1 - \text{Var}(\text{residual})/\text{Var}(\text{residual} + \text{seasonal}))$. It is not the fraction of all observed variance explained. No training values needed interpolation. STL is descriptive and supplies no model inputs. ADF/STL outputs were recomputed; strong seasonality and rejection of a unit root are compatible.

5. Methodology

A target $x(t+1)$ uses $x(t-L+1)$, ..., $x(t)$, with no future input. A fully observed 144-step history and finite target define eligibility for every candidate, including the earlier $L=36$ experiments. Missing-valued windows are excluded; an incomplete or unordered timestamp grid is rejected. The same mask prevents differing evaluation subsets. Each selected area has 5,328 training and 1,008 validation/test targets.

Table 3. Target boundaries in Europe/Rome; end dates are inclusive.

Split	Dates	Use
Training	Nov 1-Dec 8	Weights and scaling
Validation	Dec 9-15	Settings and stopping
Evaluation	Dec 16-22	Reported rolling one-step errors

Per-area training mean μ and population standard deviation σ standardize history to an $(L,1)$ sequence. RidgeAR flattens it to L lag features. All models learn $\delta = (x(t+1) - x(t)) / \sigma$. Reconstruction is $\max(0, x(t) + \sigma * \text{predicted_}\delta)$. Scaling uses only training observations; no target is imputed. Shuffling already formed training windows does not introduce future values. Test history can include newly observed test values for later one-step targets; this is not seven-day recursive forecasting.

Table 4. Final model structures.

Model	Lookback	Structure	Parameters
RidgeAR	144	L2 $\alpha=10$; fitted intercept	145
LSTM	144	16 gated units; final hidden state	1169
CausalCNN	144	16 filters; kernel 3; dilations 1,2,4,8,16,32,64	4785

Ridge minimizes squared correction error plus an L2 coefficient penalty, so regularization favors persistence. LSTM uses sigmoid gates and tanh cell updates with a linear output head, no dropout and no state carried between windows. CausalCNN uses causal ReLU convolutions and a scalar head at the final position. Its theoretical receptive field is 255; only 144 observations are available. Removing dilation 64 reduces the field to 127 and loses access to the oldest 17 observations, so trimming is an ablation, not free removal of unused history.

Neural heads start at zero. Adam uses learning rate 0.001, gradient-norm clipping 1 and batch size 128. Early stopping monitors standardized validation MSE, patience 4, $\min_delta=1e-5$, with an 80-epoch cap. Saved metadata distinguishes the restored epoch from the raw minimum-loss epoch. Weights and scaling are fitted separately per area.

The correction target does not force zero changes: minimizing squared error in δ is equivalent, up to scale, to squared forecast error after reconstruction before clipping. Limited inputs, model capacity, regularization and optimization can affect predictions. Their separate effects require controlled comparisons.

6. Iterative experimentation and hyperparameter tuning

Table 5. Area 5161, seed 42 validation experiments. These are measured runs, not equal compute budgets.

Experiment	Model	Val RMSE	Epochs	Train seconds
initial	RidgeAR	159.07	0	0.021
initial	LSTM	163.30	13	4.105
initial	CausalCNN	161.02	20	4.514
daily_history	RidgeAR	153.13	0	0.051
daily_history	LSTM	160.26	19	16.879
daily_history	CausalCNN	160.28	16	10.198
capacity	RidgeAR	152.39	0	0.041
capacity	LSTM	162.04	9	13.023
capacity	CausalCNN	160.09	11	9.126
budget	RidgeAR	152.39	0	0.040
budget	LSTM	160.26	19	17.367
budget	CausalCNN	160.09	11	9.251

Round 2 improved all models by extending six-hour history to a day. CNN dilation depth also changed, so its improvement cannot be attributed to history alone. Round 3 retained Ridge alpha 10 and LSTM width 16; CNN width 16 reduced RMSE from 160.28 to 160.09. That is a 0.12% difference. It decided the choice only because I had committed in advance to taking the lowest validation RMSE; it is far too small to claim the wider CNN is genuinely better. Each candidate used one tuning seed.

The original 20-epoch study had already produced test results when a stopping-history audit motivated round 4. Of 30 neural fits, 22 ran 20 epochs; one also triggered patience there, leaving 21 cap-only terminations. Ten had minimum recorded validation loss at the final epoch. The cap did not bind for area 5161 seed 42, but did for its CNN seeds 43 and 44. Increasing it uniformly to 80 is a budget-sensitivity revision, not evidence that finite budgets are inherently incorrect.

All 30 revised neural fits end through patience; the longest runs 55 epochs. No test target enters training or early-stopping loss, but the revised week is not a fresh untouched holdout. Original results remain in commit a5e5111; `revision_provenance.json` records this chronology. Seeds 42-44 are retained, with 42 as the reference convention; available commits do not independently establish when that convention was first chosen.

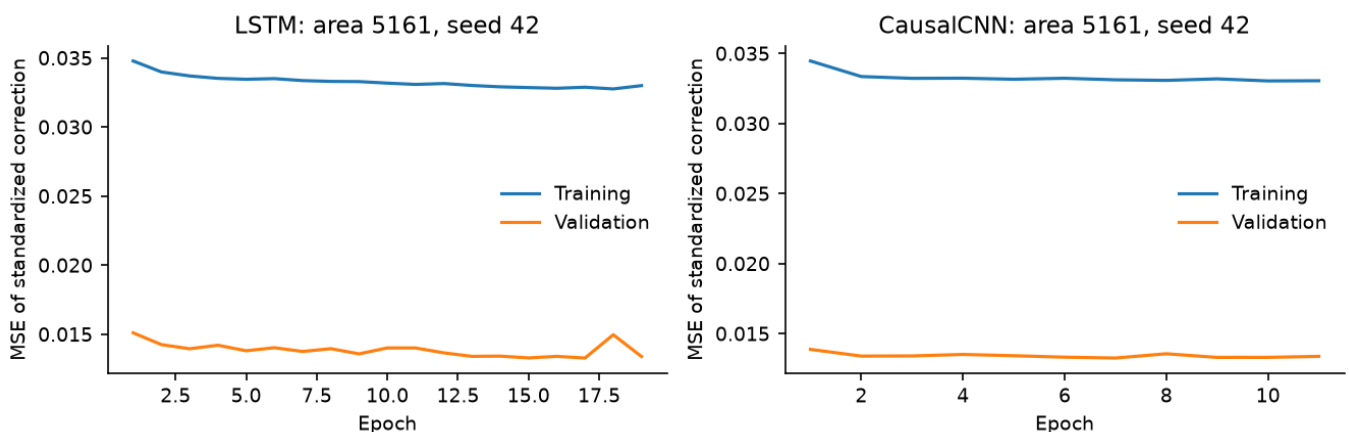


Figure 5. Training and validation correction MSE for area 5161, seed 42. Different chronological periods need not have equal loss levels.

Validation below training loss can reflect different traffic regimes and training-loss averaging during updates. It is not by itself evidence of leakage or proof of convergence.

7. Results and timing

Table 6. Area 5161; seed 42, 1,008 eligible targets.

Model (seed 42)	MAE	RMSE	MAPE (%)
RidgeAR	85.08	124.18	9.80
LSTM	81.13	118.42	9.09
CausalCNN	79.96	118.16	8.36
Persistence	92.80	134.88	9.19
Daily seasonal	338.59	619.04	25.94

Table 7. Area 5059; seed 42, 1,008 eligible targets.

Model (seed 42)	MAE	RMSE	MAPE (%)
RidgeAR	71.79	99.66	7.91
LSTM	75.46	106.22	7.78
CausalCNN	73.96	103.49	7.61
Persistence	81.52	114.38	7.96
Daily seasonal	171.74	245.87	18.02

Table 8. Area 5259; seed 42, 1,008 eligible targets.

Model (seed 42)	MAE	RMSE	MAPE (%)
RidgeAR	65.86	93.17	7.74
LSTM	66.84	95.02	7.82
CausalCNN	66.64	93.61	8.21
Persistence	75.97	109.58	8.11
Daily seasonal	470.32	861.62	71.62

MAE and RMSE use every eligible target. MAPE is 100 times mean absolute relative error over nonzero targets, with excluded counts recorded; an all-zero target set yields undefined MAPE. No epsilon is inserted. All five actual evaluation series have 1,008 scored targets and no zero targets. Activity-unit errors are reported after inverse transformation.

Table 9. Mean timing across five areas and three fits per model.

Model	Training (s)	Test batch (ms)	Single (ms)
CausalCNN	17.090	30.70	0.298
LSTM	23.126	34.59	2.067
RidgeAR	0.046	0.10	0.047

Apple M2 Pro, 16 GB RAM, CPU execution, four TensorFlow intra-op threads. Neural training includes construction, compilation and epoch-wise validation; Ridge includes fitting only. These scopes differ. Warm synchronous inference is the median of five full-batch and 30 single-sample calls per fit, then averaged. Windows, scaling and correction reconstruction are excluded. Measurements include local system noise and are not service-latency guarantees.

Forecasts for area 5161

Observed and predicted activity use identical timestamps, reference seed 42 and already observed history.

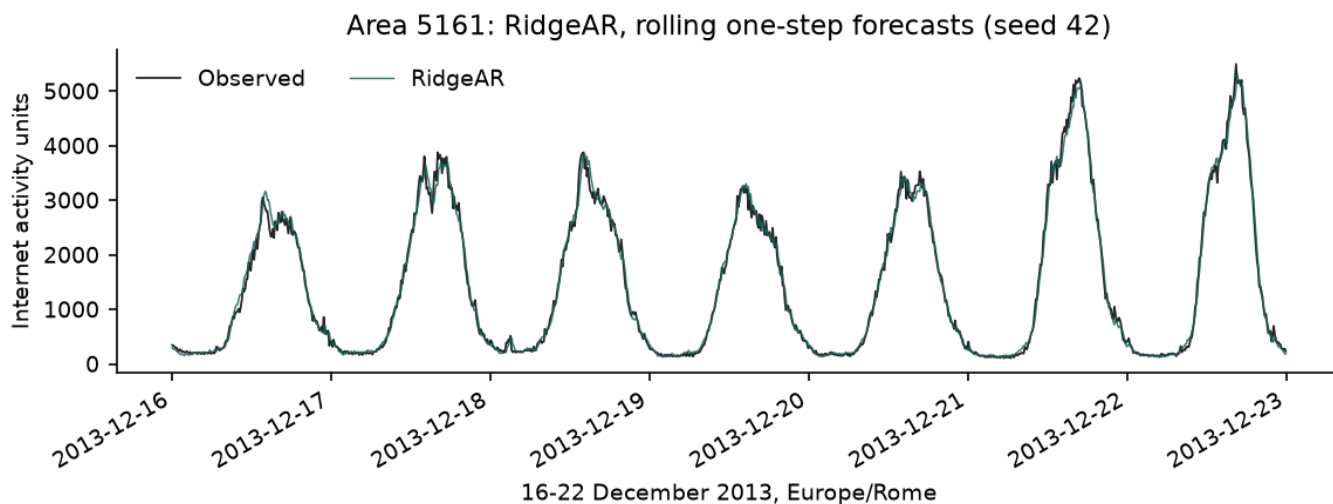


Figure 6a. RidgeAR, area 5161: December 16-22 rolling one-step forecasts.

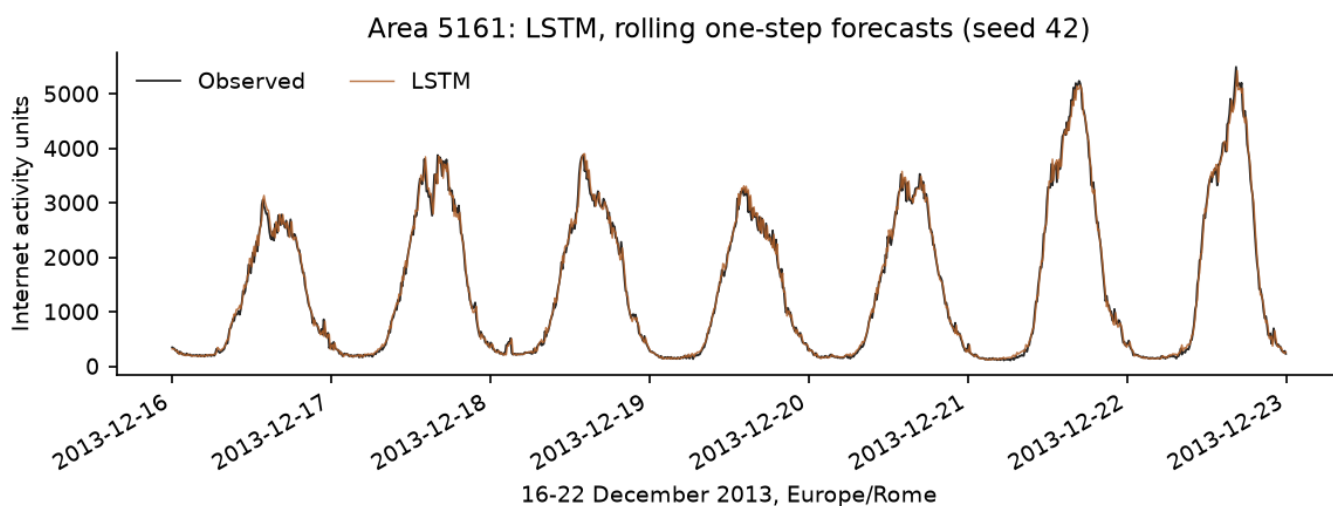


Figure 6b. LSTM, area 5161: December 16-22 rolling one-step forecasts.

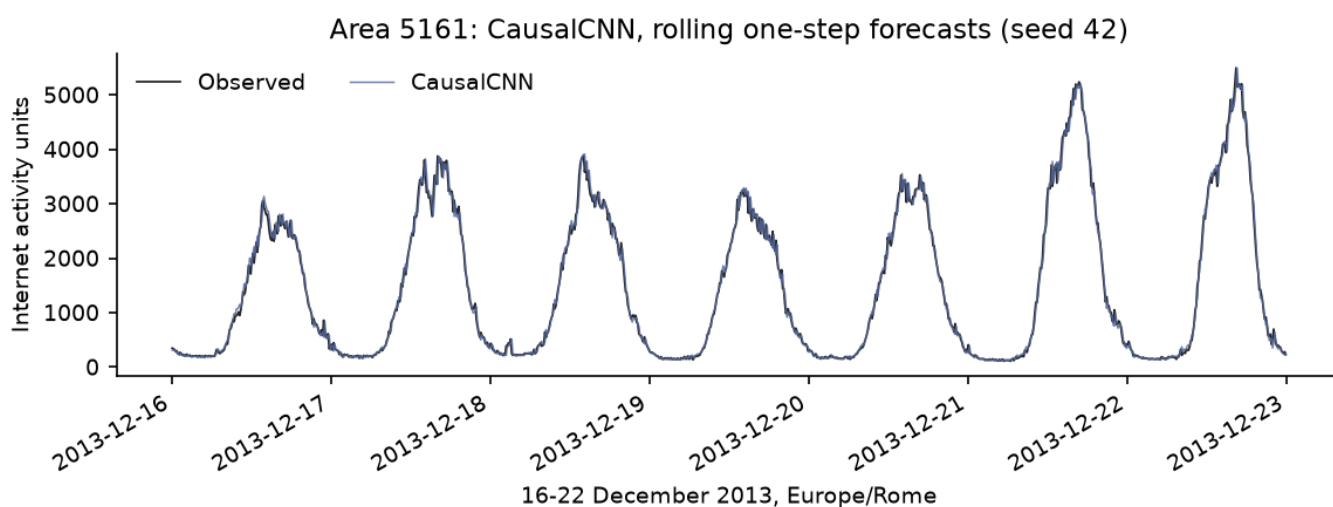


Figure 6c. CausalCNN, area 5161: December 16-22 rolling one-step forecasts.

Forecasts for area 5059

Observed and predicted activity use identical timestamps, reference seed 42 and already observed history.

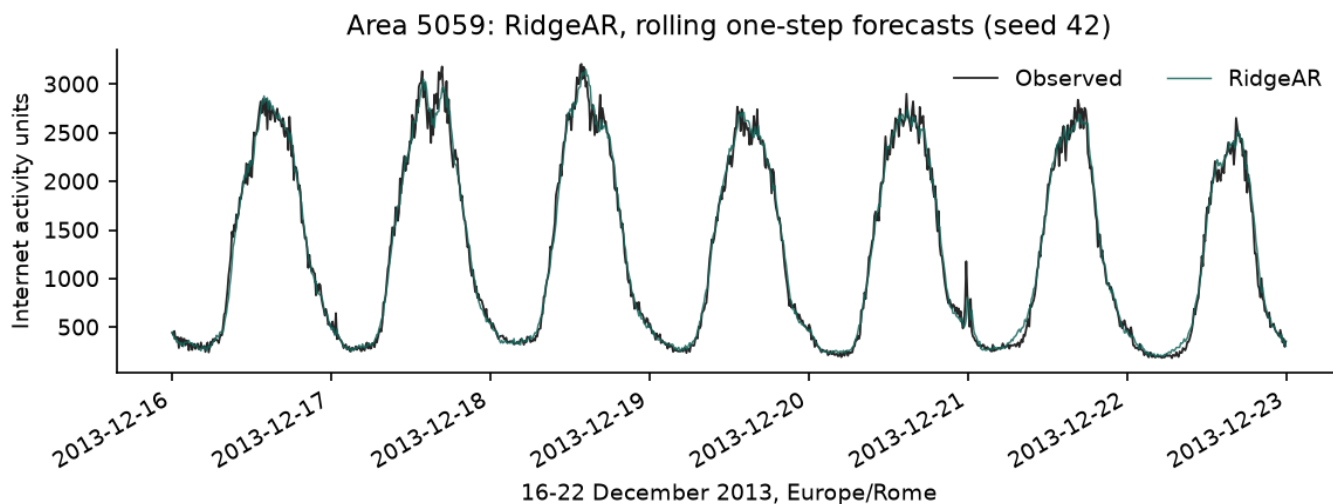


Figure 7a. RidgeAR, area 5059: December 16-22 rolling one-step forecasts.

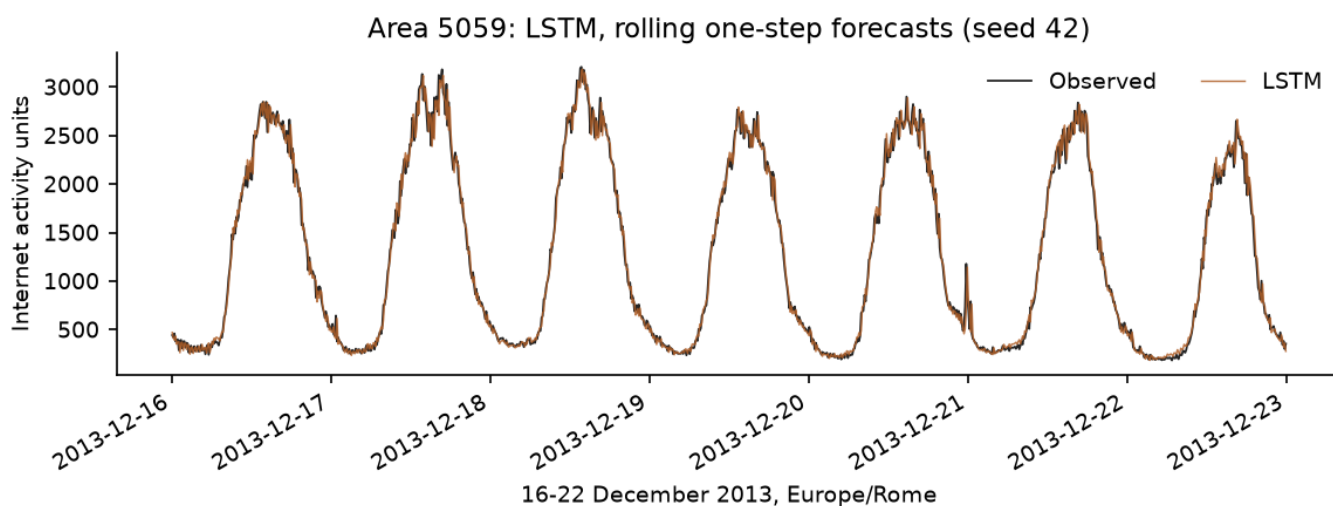


Figure 7b. LSTM, area 5059: December 16-22 rolling one-step forecasts.

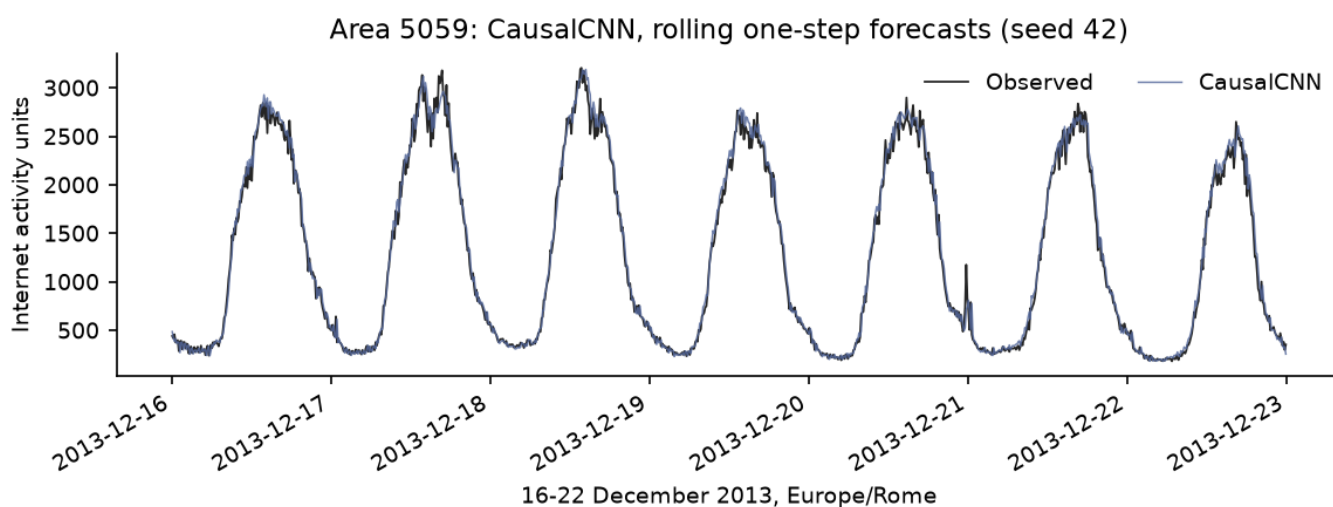


Figure 7c. CausalCNN, area 5059: December 16-22 rolling one-step forecasts.

Forecasts for area 5259

Observed and predicted activity use identical timestamps, reference seed 42 and already observed history.

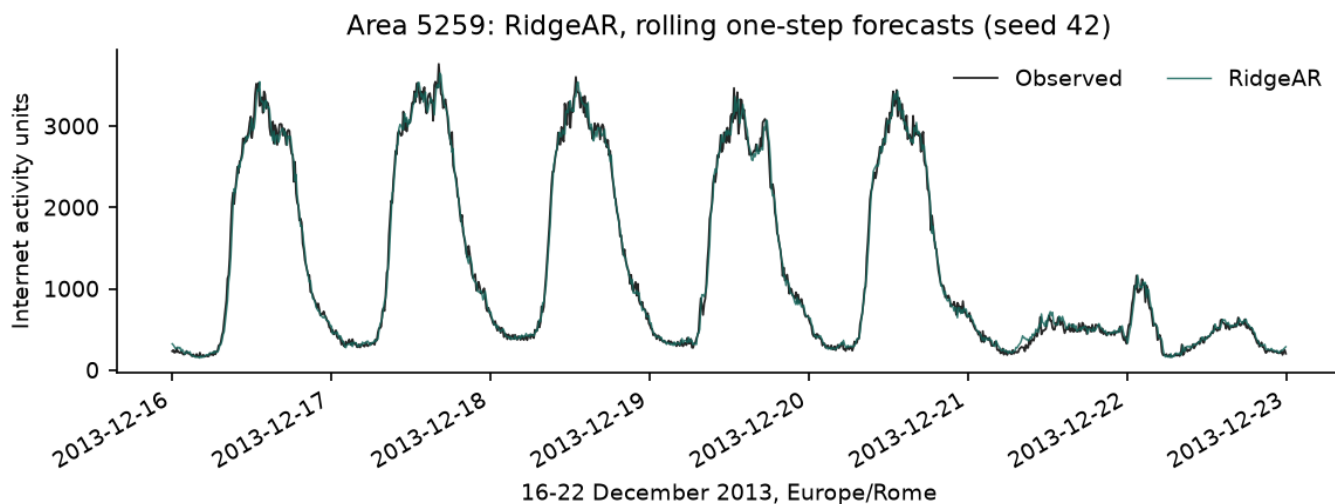


Figure 8a. RidgeAR, area 5259: December 16-22 rolling one-step forecasts.

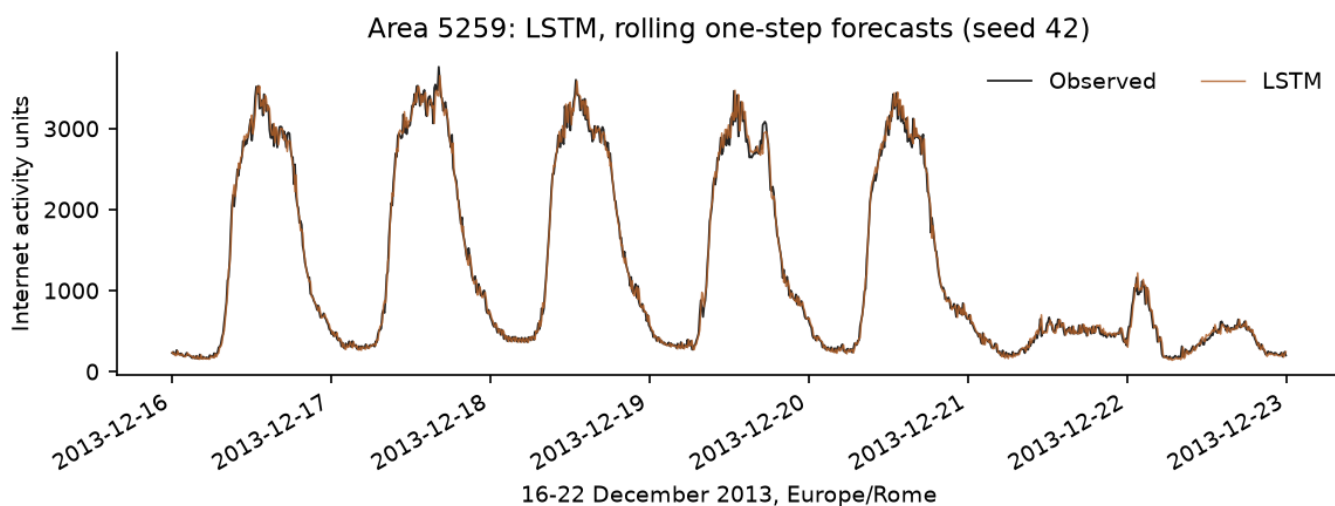


Figure 8b. LSTM, area 5259: December 16-22 rolling one-step forecasts.

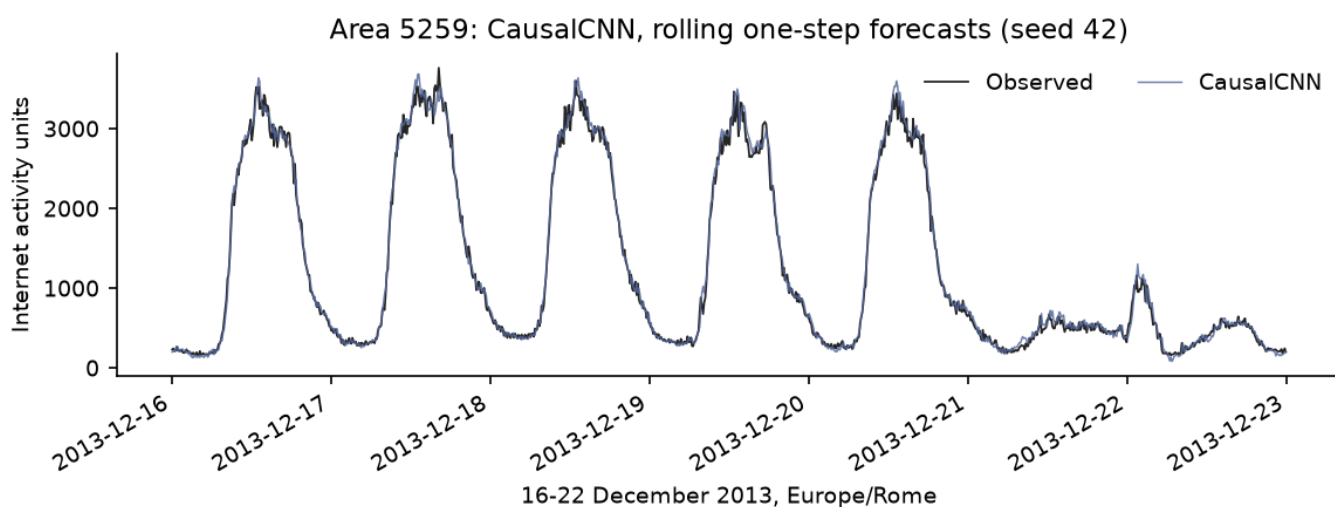


Figure 8c. CausalCNN, area 5259: December 16-22 rolling one-step forecasts.

8. Comparative discussion

Fourteen of fifteen reference-seed learned fits have lower RMSE than persistence. Area 4556 LSTM is slightly worse: 39.76 versus 39.62, about 0.35%; that is a small loss on this particular week, not a difference I have shown to be real. Daily-seasonal persistence is worse in all five areas. I read this as confirming that the immediate previous value is the thing to beat at a ten-minute horizon, not as evidence that seasonal information is useless: a model given the daily shape *alongside* recent history could still use it, and I have not tested that.

Table 10. Validation/test winners at seed 42 only.

Area	Validation best	Test best
5161	RidgeAR	CausalCNN
5059	CausalCNN	RidgeAR
5259	CausalCNN	RidgeAR
4159	LSTM	RidgeAR
4556	CausalCNN	RidgeAR

The number of ranking disagreements is 5/5 for seed 42, 2/5 for seed 43 and 2/5 for seed 44. On area 5161, CNN validation RMSE is 160.09, 152.02, 148.03, while Ridge remains 152.39. Thus optimization variation changes the validation winner; the reference-seed reversal cannot be attributed solely to a difference between weeks. Three seeds do not measure uncertainty across future weeks.

Table 11. Evaluation RMSE mean (sample SD), three seeds.

Area	RidgeAR	LSTM	CausalCNN
5161	124.18 (0.00)	120.12 (1.93)	118.49 (2.91)
5059	99.66 (0.00)	101.82 (4.43)	103.68 (2.58)
5259	93.17 (0.00)	96.05 (1.67)	91.18 (2.28)
4159	19.88 (0.00)	20.72 (1.26)	20.15 (0.78)
4556	35.57 (0.00)	42.08 (2.01)	37.65 (1.15)

On area 5259, Ridge wins at seed 42, but CNN has the lower three-seed mean. On area 5059, the lower training CV and stronger daily dependence coincide with competitive linear forecasts; five selected areas do not establish a causal link between traffic characteristics and architecture ranking.

Table 12. Area 5161 signed error by realized-target decile; 101 observations per extreme decile.

Model	Low-decile bias	High-decile bias	High-decile MAE
RidgeAR	+16.4	-80.8	164.4
LSTM	+13.3	-50.3	151.8
CausalCNN	+4.3	-36.2	152.1

Across areas, 14/15 high-decile biases are negative and 13/15 low-decile biases positive. I should be careful about what this proves. The deciles are defined by what actually happened, so even a perfect forecast would look biased this way at the extremes: the unpredictable part of a spike can only land on one side. The pattern is a useful description of where the errors are worst, but it does not prove the correction target is what causes it. Training a model to predict the value directly instead would test that.

8.1 Failure case and limits of the explanation

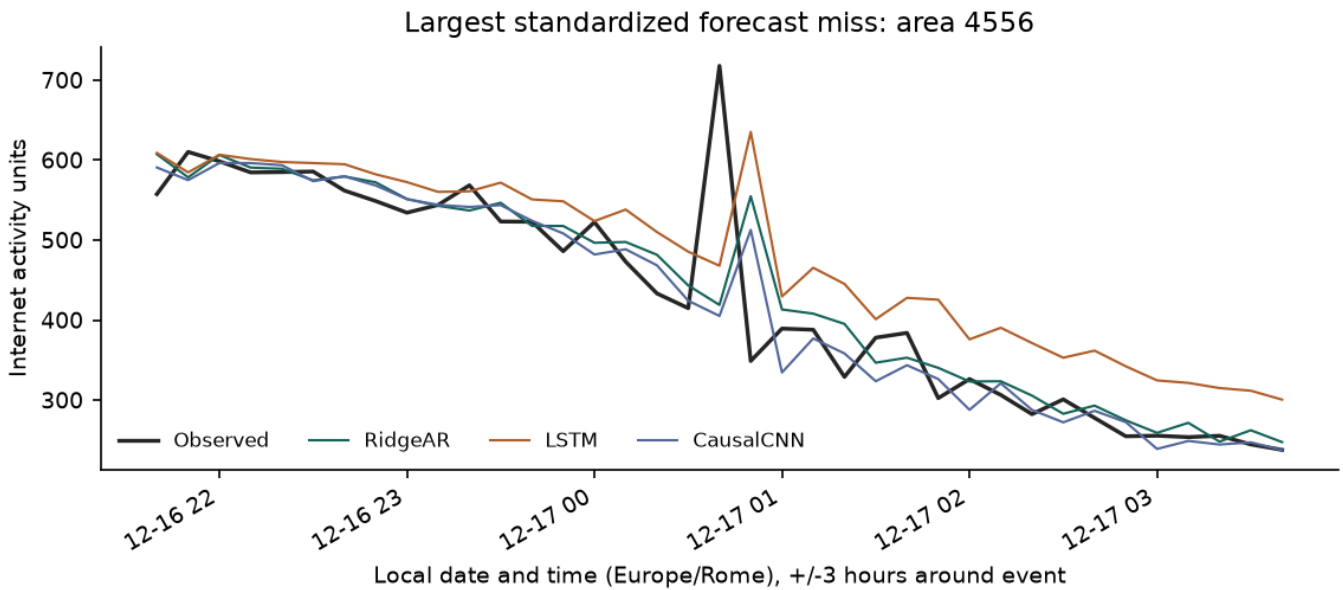


Figure 9. The biggest single error at seed 42, scaled by each area's training standard deviation so areas of different size compare fairly. I picked this case after the models were finished.

The selected miss is area 4556 at 2013-12-17 00:40 CET. CausalCNN predicts 405.11 against 717.77, an absolute error of 312.66, or 1.24 training SDs. Activity rises from 414.87 to 717.77 and then falls to 348.86. All three models miss the rise and overshoot the fall. The following interval has actual 348.86, compared with RidgeAR 554.76, LSTM 635.22, CausalCNN 512.68.

The spike enters the next window and persistence anchor, which is consistent with the observed lagging response. A model could in principle learn to cancel that, so I would not claim this is a hard limit of the architecture or that more tuning could not help. This is also the single worst error I could find, so it is deliberately not a typical case. I have no evidence of what caused the spike, and with one interval involved I cannot rule out a measurement glitch.

A defensible next experiment would compare direct-target and correction-target models under matched validation and compute protocols, then assess abrupt changes on additional periods. Neighboring-area observations may contain predictive information, but that benefit has not been demonstrated. An asymmetric loss could prioritize costly underprediction, with its accuracy/calibration trade-off evaluated separately.

9. Conclusion and future work

The busiest area's reference CNN improves RMSE from persistence's 134.88 to 118.16. RidgeAR stays competitive at a fraction of the training cost. Rankings move with area, seed and week, so I am not willing to name a best architecture on this evidence. What I would defend is narrower: at this horizon a 145-parameter linear model is a serious competitor, and the burden is on the more expensive models to show a gain that survives more than one week.

The main limits are these. I have one evaluation week, and I reused it after revising the epoch budget. The areas were chosen using totals that include that week. Hyperparameters were tuned mostly on one area and one seed. Some aggregation bins are only partly covered. And the training budgets are not matched across models. What I would do next is choose the areas from an earlier period, evaluate across several weeks with a rolling origin, give each model the same compute, and test whether predicting the value directly beats predicting a correction. The daily and weekly correlation suggests calendar features might help, but that needs testing rather than assuming.

AI assistance disclosure

I used AI assistance throughout this project: writing and debugging the code, designing the experiments, building the models, writing the tests, running the analysis, making the figures, and drafting this report. It also helped me audit my own work, which is how I found the epoch-budget problem and a memory figure that did not reproduce. Every number here comes from the real dataset and can be recomputed from the saved predictions in the repository. The only made-up values anywhere are the small fixtures in the unit tests.

I am responsible for what I am submitting, including being able to explain and justify the data handling, the memory decisions, the model choices, the experiments and the conclusions I draw from them.

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